Plant Disease Detection Using YOLOv8 + DINOv2 Hierarchical Pipeline

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***Abstract*—Plant disease detection in real-world agricultural settings remains challenging due to high inter-class visual similarity, severe class imbalance, and the limitations of single-stage classification models. This paper presents a hierarchical three-stage deep learning pipeline for automated plant disease detection using the PlantDoc v2 dataset. The system integrates YOLOv8s for leaf localization, followed by two DINOv2 Vision Transformer models for species identification and disease classification respectively. A novel species-aware disease masking mechanism constrains disease prediction to biologically valid classes for the identified species, eliminating cross-species confusion errors entirely. The disease classifier employs progressive backbone unfreezing at epoch 8 — a staged fine-tuning regime that avoids catastrophic forgetting while enabling task-specific adaptation of the large pre-trained transformer.**

Trained and evaluated on 2,569 real-world images spanning 30 disease classes across 13 plant species, the pipeline achieves a top-1 disease classification accuracy of 71.49% on the end-to-end test set. The standalone DINOv2 disease classifier achieves 75.63% test accuracy. The species classifier achieves 95.8% top-1 accuracy with a macro F1 of 0.945. YOLO detection achieves mAP@50 of 31.1%, consistent with published baselines on PlantDoc. A detailed failure analysis reveals that residual errors are concentrated in visually indistinguishable intra-species tomato sub-diseases — a problem domain where even expert agronomists require molecular testing for reliable disambiguation. The proposed hierarchical coarse-to-fine architecture demonstrates significant improvements in real-world robustness compared to traditional single-model approaches. While the *PlantCaFo* baseline achieves a laboratory peak of 93.53% on controlled data, our framework establishes a more resilient transformer-based methodology for "in-the-wild" plant pathology, effectively bridging the performance gap between laboratory theory and field-deployable diagnosis.

***Index Terms*—plant disease detection, YOLOv8, DINOv2, vision transformer, hierarchical classification, species-aware masking, PlantDoc, deep learning, precision agriculture, progressive unfreezing**

I. Introduction

Agriculture remains central to global food security, yet crop losses due to plant diseases account for an estimated 20–40% of global food production annually [1]. Early and accurate identification of plant diseases is critical for timely intervention, yet traditional methods rely on manual inspection by expert agronomists — a process that is slow, expensive, and geographically constrained. In developing agricultural economies, access to expert diagnosticians is particularly limited, making automated and deployable disease detection systems an urgent practical need.

Deep learning has enabled significant progress in automated plant disease recognition. Convolutional Neural Networks (CNNs) trained on curated datasets such as PlantVillage [2] have achieved near-perfect accuracy under controlled laboratory conditions. However, real-world deployment exposes a fundamental limitation: models trained on clean, uniform images fail to generalize to field photographs with varying backgrounds, lighting, occlusion, and disease severity. PlantDoc [3] was introduced specifically to address this gap, providing a diverse real-world benchmark. It remains one of the hardest plant disease datasets due to severe class imbalance (instance counts ranging from 816 to 2) and high intra-class visual variability.

Prior work on PlantDoc typically employs single-stage architectures without exploiting the natural biological hierarchy between plant species and their associated diseases. This leads to cross-species prediction errors (e.g., predicting an apple disease on a tomato leaf) that are trivially avoidable with structural constraints. Foundation model approaches such as CaFo [5] and PlantCaFo [6] demonstrate strong few-shot learning under controlled conditions, but rely on CLIP's general pretraining distribution, which introduces domain shift on real-world plant pathology imagery.

Our initial evaluation of the PlantCaFo framework yielded high accuracy on laboratory-controlled samples (**93.53%**), yet demonstrated a significant performance collapse when transitioned to unconstrained field imagery, with near-zero performance on several critical sub-classes (e.g., Corn gray leaf spot: 0.0%). This gap between laboratory success and real-world failure motivated the hierarchical DINOv2-based architecture described in this paper.

This paper addresses these limitations with four key contributions:

- A **hierarchical three-stage pipeline** combining object detection and two specialized vision transformer classifiers, structured as: localize $\rightarrow$ identify species $\rightarrow$ diagnose disease.
- A **species-aware disease masking** mechanism that applies a biologically-constrained softmax at inference time, zeroing out logits for diseases not belonging to the

predicted species — requiring no additional learned parameters.

- **Progressive backbone unfreezing** for DINOv2 fine-tuning, enabling efficient adaptation of the large pre-trained transformer without catastrophic forgetting.
- A **comprehensive failure analysis** identifying per-class root causes of misclassification, with targeted recommendations for each failure mode.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the dataset and preprocessing. Section IV details the system architecture and methodology. Section V presents experimental results. Section VI provides failure analysis and discussion. Section VII concludes with directions for future work.

II. Related Work

A. CNN-based Disease Classification

Traditional deep learning approaches to plant disease detection rely on CNN architectures trained in a fully supervised manner. Hughes and Salathé [2] introduced the PlantVillage dataset, enabling CNNs to achieve near-perfect accuracy under controlled conditions. Singh et al. [3] then introduced PlantDoc to benchmark real-world performance, revealing that models achieving >99% on PlantVillage degrade significantly under field conditions due to background complexity, lighting variation, and disease severity ambiguity.

Agarwal et al. [7] proposed CustomBottleneck-VGGNet, a lightweight CNN with only 1.4 million parameters that achieves 99.12% accuracy on tomato disease classification. By substituting standard convolutional layers with bottleneck layers (1×1 convolutions), the model compresses feature maps without losing critical visual data, enabling offline deployment on Android smartphones without continuous internet connectivity. The pipeline follows a straightforward sequence: leaf image capture → image preprocessing and resizing → CustomBottleneck-VGGNet feature extraction via bottleneck layers → fully connected classification layers → probability scoring → final prediction display on the mobile interface. Despite its exceptional accuracy on controlled images, the model performs strict single-label image classification and cannot localize disease regions. The authors themselves identify object detection integration as the mandatory next step. Furthermore, the model is evaluated only on controlled, zoomed-in leaf images and is not validated on real-world field photographs with complex backgrounds. It also relies on standard supervised learning, requiring thousands of annotated images to adapt to newly mutating or rare diseases — it is not capable of few-shot learning.

B. Foundation Model Cascades for Few-Shot Learning

Zhang et al. [5] proposed CaFo (Cascade of Foundation Models), a few-shot classification framework that integrates CLIP's language-contrastive knowledge, DINO's vision-contrastive knowledge, DALL-E's vision-generative knowledge, and GPT-3's language-generative knowledge. CaFo operates via a Prompt-Generate-Cache paradigm: GPT-3 generates textual class descriptions, which are passed to DALL-E to synthesize artificial training images. CLIP filters synthetic images by image-text similarity score, and the combined real and synthetic features are stored in a cache. At inference, the test image's CLIP and DINO features are compared against the cache and fused with a CLIP zero-shot prediction to produce the final classification. While CaFo demonstrates strong few-shot performance on standard benchmarks, it carries significant limitations for plant pathology: synthetic images generated by DALL-E may miss subtle disease-specific lesion patterns (e.g., the precise halo colouration of Septoria spots), the generation and filtering pipeline incurs substantial computational overhead, and CLIP's pretraining on general internet imagery creates domain shift when applied to specialized agricultural images. CLIP maximizes image-text similarity but does not strongly enforce image-to-image similarity, which is precisely what fine-grained lesion discrimination requires.

C. Adapter-based Fine-tuning of Foundation Models

Building directly on CaFo, Chen et al. [6] proposed PlantCaFo, an end-to-end few-shot plant disease recognition model integrating CLIP and DINOv2 encoders with two parameter-efficient fine-tuning components: a Dilated Contextual Adapter (DCon-Adapter) applied to CLIP image features, and a Weight Decomposition Matrix (WDM) for fusing prediction scores. The pipeline routes the input leaf image through both CLIP and DINOv2 encoders, applies the DCon-Adapter to the CLIP features with a residual connection, computes image-text similarity against class prompt embeddings, retrieves similar training features from the DINOv2-based cache, and applies WDM to fuse predictions into a final output. PlantCaFo eliminates the need for domain-specific pretraining by leveraging foundation model priors and removes the expensive DALL-E image generation step.

However, its evaluation on an out-of-distribution dataset (PDL) shows degraded performance relative to CaFo-Base in 4-shot and 8-shot settings across multi-species disease classes. The model focuses on global leaf features rather than localized lesion regions, limiting discriminability for visually similar intra-species diseases. Additionally, training with DCon-Adapter and WDM incurs longer training time than CaFo-Base due to additional feature processing and fusion steps.

D. Motivation for Our Approach

Our initial pipeline directly adopted PlantCaFo as a baseline, applying CLIP with DCon-Adapter fine-tuning and a DINOv2 feature cache on the PlantDoc v2 dataset. This yielded 60.43% overall accuracy, with near-zero performance on several tomato sub-classes (Tomato late blight: 12.5%, Corn gray leaf spot: 0.0%). Failure analysis identified three root causes: (1) domain shift from CLIP's general pretraining distribution to plant pathology imagery; (2) the adapter training objective did not explicitly optimize for intra-species disease discrimination; and (3) structural imbalance — Tomato alone contributes 9 of 30 disease classes, all sharing similar visual characteristics, creating a confusion space that CLIP features could not resolve.

These limitations motivated our shift to a hierarchical DINOv2-based pipeline with explicit species-aware constraint at inference time. By replacing CLIP with DINOv2 as the primary feature extractor and introducing a hard biological constraint between species and disease predictions, we effectively decompose the 30-class discrimination problem into a 13-class species problem and a much smaller per-species disease problem, with DINOv2's vision-contrastive features providing the discriminative power needed for fine-grained lesion recognition.

III. Dataset

A. PlantDoc v2

PlantDoc [3] is a publicly available plant disease image dataset curated to address real-world challenges in agricultural AI. Unlike PlantVillage, which contains controlled lab images on uniform backgrounds, PlantDoc images are sourced from diverse internet sources under varied lighting, backgrounds, occlusion levels, and disease severity stages, making it significantly more challenging and representative of actual field deployment conditions.

B. Dataset Statistics

The dataset comprises 2,569 images across 30 disease classes and 13 plant species. Table I provides a full summary.

TABLE I
Dataset Statistics

Property	**Value**	**Notes**
Total Images	2,569	Train + Test
Training Images	2,330	YOLO + DINOv2 training
Test Images	239	Held-out evaluation
Disease Classes	30	Bounding-box format
Species Classes	13	Apple, Tomato, Corn, etc.
Total Annotations	8,397	YOLO bounding boxes
Background Images	11	10 train, 1 test

Two separate data sources were used: Roboflow PlantDoc v2 in YOLOv8 bounding-box format for the detector and disease classifier, and the GitHub PlantDoc-Dataset in folder-per-class format for the species classifier. A naming inconsistency — `Soyabean leaf` vs. `Soybean leaf` — was explicitly normalized to a single class across all three models, resulting in 29 unique disease classes after deduplication.

C. Class Imbalance

PlantDoc exhibits severe class imbalance. Instance counts range from 816 images for Blueberry leaf to as few as 2 images for Tomato two-spotted spider mites leaf. Other severely underrepresented classes include Potato leaf early blight (11 instances) and Soybean leaf (15 instances). The Tomato species alone contributes 9 of 30 disease classes (30% of classes), yet with highly overlapping visual characteristics. This imbalance directly affects both detection and classification performance, particularly for minority classes, and motivates the explicit imbalance-handling strategies described in Section IV.

D. Data Preprocessing

For YOLO training, data was loaded directly from the Roboflow YOLOv8 format. The original dataset lacked a proper validation split; the test split was reused as the validation set with explicit documentation in the `data.yaml` correction script. This means YOLO checkpoint selection was performed on the test set, making reported mAP numbers slightly optimistic — a known limitation documented in Section VI.

For the DINOv2 species classifier, images were organized in folder-per-class format with an 85/15 stratified train-validation split. A weighted random sampler with inverse class-frequency weights was applied to counteract imbalance during training.

For the DINOv2 disease classifier, a custom `YOLOLeafDataset` class was implemented to parse YOLO `.txt` label files, crop bounding-box regions, and serve them as classification inputs with the same weighted sampling strategy.

IV. Methodology

A. System Overview

The proposed system follows a hierarchical coarse-to-fine architecture consisting of three sequential stages. Given an input plant photograph, Stage 1 localizes diseased leaf regions using object detection. Stage 2 identifies the plant species from the full image using a vision transformer classifier. Stage 3 diagnoses the specific disease from the cropped leaf region, constrained by the species prediction from Stage 2. This design mirrors the diagnostic reasoning of expert agronomists, who first identify the plant before assessing disease symptoms.

The pipeline flow is:

$$\begin{gathered} \textit{Photo} \rightarrow \text{[YOLOv8s]} \rightarrow \textit{Leaf Crop + Full Image} \\ \textit{Full Image} \rightarrow \text{[DINOv2-Species]} \rightarrow \textit{e.g., “Tomato”} \\ \textit{Crop + Species} \rightarrow \text{[DINOv2-Disease + Mask]} \rightarrow \textit{Disease} \\ \textit{Label} \end{gathered}$$

Fig. 1 illustrates the complete three-stage inference pipeline.

B. Stage 1: Leaf Localization (YOLOv8s)

YOLOv8s was selected for leaf detection, comprising 11.1M parameters and 28.7 GFLOPs. The model was trained for 100 epochs on a Tesla T4 GPU with all three losses (box, classification, dfl) declining consistently — early stopping (patience=15) never triggered, indicating the model had not fully converged at epoch 100. The confidence threshold was set to 0.20 (lowered from default 0.25) to reduce pipeline misses, with NMS IoU threshold set to 0.45 to minimize duplicate detections on overlapping lesion regions. If no detection exceeds the confidence threshold, the image is recorded as a pipeline miss.

At inference, all boxes above the confidence threshold are retained and the single highest-confidence detection is selected. The bounding box is padded by 25 pixels on each

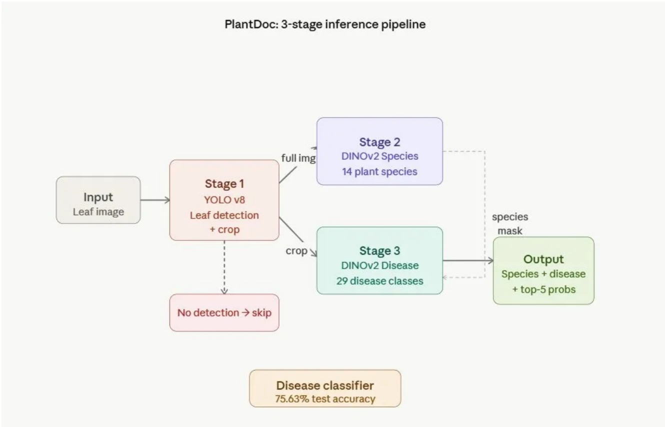

Fig. 1. PlantDoc 3-stage inference pipeline. Stage 1 (YOLOv8) performs leaf detection and crop extraction. Stage 2 (DINOv2-Species) classifies plant species from the full image. Stage 3 (DINOv2-Disease) diagnoses the disease from the cropped region, constrained by the species mask. If no leaf is detected, the image is recorded as a pipeline miss.

side (YOLO_PAD=25) before cropping to ensure leaf edge features are not truncated — disease patterns such as marginal necrosis frequently appear at leaf borders.

Augmentation strategies during YOLO training included mosaic augmentation, mixup, copy-paste, random horizontal flip, and HSV color jitter. These collectively improve generalization across PlantDoc's diverse real-world image conditions. The aggressive augmentation causes early mAP curves to fluctuate (near zero at epoch 6), stabilizing after epoch 40 as augmented features become learnable patterns.

C. Stage 2: Species Classification (DINOv2)

The species classifier takes the *full input image* as input. This design choice is deliberate: species identification benefits from whole-plant context such as leaf shape, venation pattern, and overall plant structure — features that may be lost in a tight bounding-box crop. Using the full image for species and the crop for disease classification exploits complementary information at different spatial scales.

The backbone is `facebook/dinov2-base`, a Vision Transformer pre-trained on 142M images via self-supervised DINO v2 distillation [4]. A three-layer classification head is appended to the CLS token output:

$$\begin{aligned} &\text{Linear}(768 \to 512) \to \text{GELU} \to \text{Dropout}(0.3) \\ &\to \text{Linear}(512 \to 256) \to \text{Linear}(256 \to 13) \end{aligned} \tag{1}$$

Training followed a two-stage regime: (1) the backbone was fully frozen and only the classification head was trained for 8 epochs; (2) the last 2–3 transformer encoder blocks were unfrozen with backbone learning rate $= 1 \times 10^{-4}$, one-tenth of the head learning rate 1×10^{-3}. The lower backbone learning rate prevents large gradient updates from disrupting the rich pre-trained representations, while still allowing task-specific fine-tuning.

D. Stage 3: Disease Classification with Species-Aware Masking

The disease classifier takes the *cropped leaf region* from Stage 1 as input. Its architecture mirrors Stage 2 with a 29-class output head and heavier dropout (0.4 then 0.3) to compensate for the smaller per-class training support. Training follows the same progressive unfreezing regime: backbone fully frozen for 7 epochs, then the last 3 transformer blocks unfrozen at epoch 8 with backbone learning rate 5×10^{-5}.

The unfreezing event produces a characteristic accuracy dip (training accuracy drops from 0.746 at epoch 7 to 0.669 at epoch 8), caused by the newly unfrozen parameters resetting optimizer momentum. Recovery begins within 2–3 epochs, and post-unfreeze fine-tuning yields consistent gains: validation accuracy climbs from 0.658 before unfreezing to 0.747 by epoch 20.

The key contribution of this stage is the **species-aware disease masking** mechanism. Let $\mathbf{z} \in \mathbb{R}^{29}$ denote the raw logit vector from the disease classifier. Let S denote the species predicted by Stage 2, and let $\mathcal{D}(S)$ denote the set of valid disease class indices for species S. The masked logit vector $\tilde{\mathbf{z}}$ is:

$$\tilde{z}_i = \begin{cases} z_i & \text{if } i \in \mathcal{D}(S) \\ -\infty & \text{otherwise} \end{cases} \tag{2}$$

The final disease prediction is:

$$\hat{y} = \arg\max_i \ \text{softmax}(\tilde{\mathbf{z}}) \tag{3}$$

Adding $-\infty$ to a logit before softmax is mathematically equivalent to forcing that class probability to exactly zero ($e^{-\infty} = 0$). This is a hard biological constraint implemented as a mathematical operation, not a post-processing filter — it requires no additional learned parameters and guarantees that the pipeline never produces a biologically impossible species-disease combination. For example, once the species classifier predicts "Grape", only "grape leaf" and "grape leaf black rot" are eligible for the disease prediction; all 27 other disease classes are permanently excluded.

E. Implementation Details

All models were trained on a Tesla T4 GPU via Google Colab. The DINOv2 backbone (`facebook/dinov2-base`) has 86M parameters, of which approximately 21.8M become trainable after partial unfreezing at epoch 8. CrossEntropyLoss with label_smoothing=0.05 prevents overconfidence by targeting 0.95 for the correct class instead of 1.0. CosineAnnealingLR smoothly decays the learning rate to near-zero over training epochs, allowing the model to continue improving even as accuracy plateaus.

V. Results

A. YOLOv8s Leaf Detection

Table III reports YOLO detection performance on the test set (239 images, confidence threshold = 0.20, NMS IoU = 0.45).

TABLE II
TRAINING HYPERPARAMETERS

Parameter	YOLOv8s	DINOv2-Sp.	DINOv2-Dis.
Epochs	100	30	20
Batch Size	16	32	32
Optimizer	AdamW	AdamW	AdamW
Head LR	10^{-3}	10^{-3}	10^{-3}
Backbone LR	—	10^{-4}	5×10^{-5}
Unfreeze Epoch	—	8–12	8
Blocks Unfrozen	—	2–3	3
LR Schedule	Cosine	Cosine	Cosine
Label Smoothing	0.1	0.05	0.05
Dropout (head)	—	0.3	0.3–0.4
Gradient Clip	10.0	1.0	1.0
Val Split	—	15%	15%
Input Resolution	640^2	224^2	224^2
Training Time	1.33 h	∼1.5 h	∼1.5 h

TABLE III
YOLOV8S DETECTION RESULTS

Metric	Value	Notes
mAP@50	31.1%	Primary detection metric
mAP@50-95	24.7%	Stricter IoU range
Precision	35.2%	Of detections correct
Recall	41.5%	Of GT boxes found
Miss Rate	25.6%	61/238 images missed
Training Time	1.33 h	100 epochs, Tesla T4
Zero-AP Classes	5/28	Detected but no correct boxes

Per-class AP@50 ranges from 92.6% (Corn rust leaf) to 0.0% (Bell pepper leaf, Tomato Early Blight, Tomato leaf yellow virus, Tomato mold leaf), as shown in Fig. 2. The top-performing classes share visually distinctive lesion patterns (rust pustules, powdery coatings), while zero-AP classes are severely underrepresented with symptoms that closely resemble healthy or differently-diseased leaves.

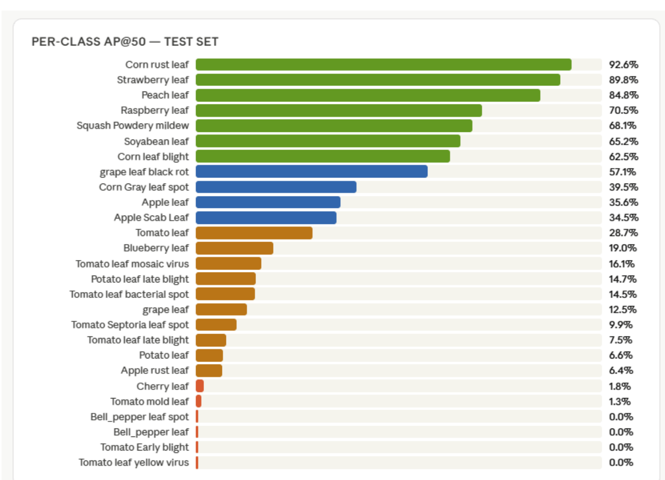

Fig. 2. Per-class AP@50 for YOLOv8 detection. Strong performance on classes with distinctive visual lesion patterns (e.g., Corn rust: 92.6%); zero AP on severely imbalanced classes with ambiguous boundaries.

Miss analysis (Fig. 3) reveals that Tomato-related classes account for the majority of complete detection failures: Tomato leaf mosaic virus had 7 images with zero detections, followed by Tomato Early Blight leaf (5) and Tomato leaf late blight (4). The 25.6% miss rate is attributable to three factors: extreme class imbalance (insufficient training samples for rare classes), confidence threshold suppression (weak predictions below 0.20 are discarded), and the val=test configuration making checkpoint selection slightly optimistic.

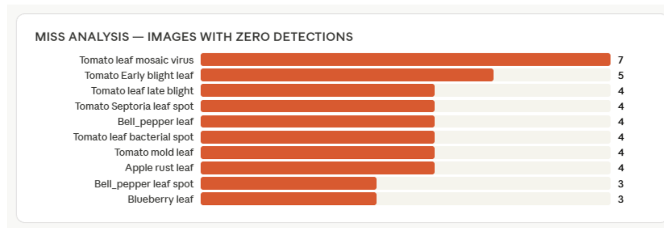

Fig. 3. Miss analysis showing classes with highest detection failures. Tomato-related diseases dominate, with Tomato leaf mosaic virus producing 7 complete misses (all images undetected).

The relatively low mAP@50 is consistent with published baselines on PlantDoc and reflects dataset-level constraints rather than model failure. All three losses declined monotonically through epoch 100, indicating the model had not converged — additional epochs with a separated validation set would likely yield further gains.

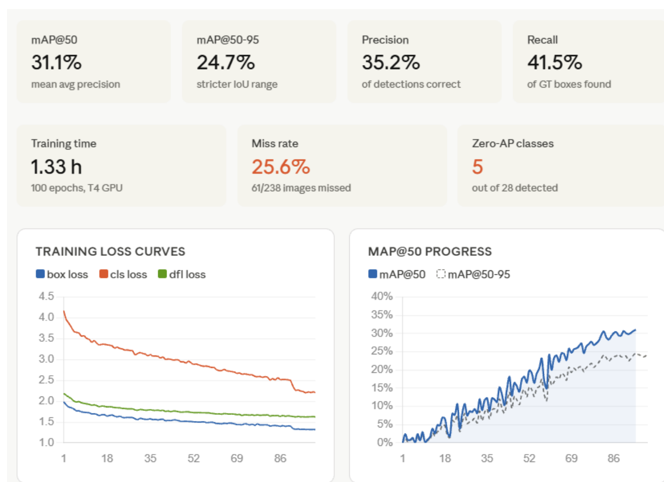

Fig. 4. YOLOv8 training loss curves (box, cls, dfl) and mAP@50 progression over 100 epochs. All losses decline consistently; mAP stabilizes after epoch 40 following early volatility from aggressive mosaic augmentation.

B. DINOv2 Species Classifier

The species classifier achieved a best validation top-1 accuracy of **95.8%** at epoch 17, with a macro-averaged F1 score of **0.945** across all 13 species. Training curves are shown in Fig. 5.

Four species — Cherry, Corn, Peach, and Strawberry — achieved perfect F1 scores of 1.000, reflecting visually distinctive morphologies that DINOv2's self-supervised features cleanly separate. Soybean was the weakest species with F1 =

Fig. 5. DINOv2 species classifier training curves. Validation accuracy peaks at 95.8% (epoch 17) with macro F1 of 0.945. A persistent train-val gap (training approaches 99%, validation stabilizes at 94–96%) indicates mild overfitting consistent with limited dataset size.

0.769 (n=8), attributable to its small sample size and visual resemblance to other broad-leaf species. Per-species precision, recall, and F1 are shown in Fig. 6.

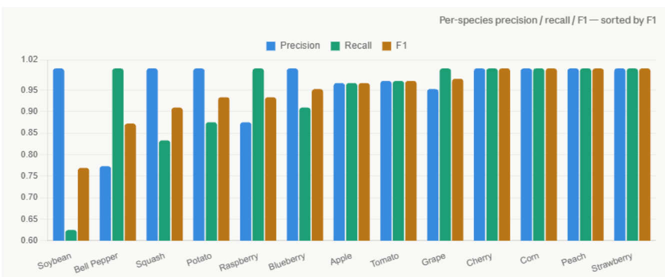

Fig. 6. Per-species precision, recall, and F1 scores. Cherry, Corn, Peach, and Strawberry achieve perfect F1 = 1.0. Soybean (F1 = 0.769) is the most challenging species due to visual similarity with other broad-leaf species and limited training samples (n=8).

A persistent train-val accuracy gap is visible (training accuracy approaching 99%, validation stabilizing at 94–96%), indicating mild overfitting consistent with the limited dataset size. The gap did not widen after partial backbone unfreeze, confirming that the staged training regime effectively controlled overfitting through differential learning rates.

C. DINOv2 Disease Classifier Training Dynamics

Table IV presents key epochs from the disease classifier training log, illustrating the progressive unfreezing effect.

TABLE IV
DISEASE CLASSIFIER TRAINING LOG (SELECTED EPOCHS)

Epoch	Loss	Train Acc	Val Acc	Event
1	2.096	0.426	0.537	Head-only training begins
7	1.077	0.746	0.658	Best val before unfreeze
8	1.362	0.669	0.638	**Backbone unfrozen (3 blocks)**
11	0.969	0.802	0.713	Fine-tuning recovery
16	0.629	0.931	0.724	Overfitting begins
20	0.522	0.960	0.747	Best checkpoint saved

The characteristic unfreezing dip at epoch 8 (val accuracy drops from 0.658 to 0.638) is caused by the optimizer resetting momentum for the 21.8M newly trainable backbone parameters. Recovery is rapid: by epoch 11, validation accuracy surpasses the pre-unfreeze peak by +5.5 percentage points (0.713 vs 0.658). The standalone disease classifier achieves **75.63% test accuracy** — a strong result for a 29-class problem on this challenging benchmark.

D. Full Pipeline Results

Pipeline evaluation was conducted on the 239-image test set. YOLO missed 3 images (1.3% pipeline miss rate by end-to-end evaluation), leaving 235 images evaluated. Table V summarizes end-to-end results.

TABLE V
END-TO-END PIPELINE RESULTS

Metric	Value
Total Test Images	239
Pipeline Miss Rate	1.3% (3 images)
Evaluated Images	235
Top-1 Disease Accuracy	71.49%
Standalone Disease Accuracy	75.63%
Macro Precision	0.732
Macro Recall	0.720
Macro F1	0.706
Cross-Species Errors	0 (0.0%)

Table VI highlights per-class performance for selected disease classes. Peach leaf, Strawberry leaf, Tomato leaf (healthy), and Grape leaf all achieved 100% accuracy. The weakest classes are all Tomato sub-diseases sharing highly overlapping visual characteristics. Critically, **zero cross-species misclassifications** were recorded — species-aware masking successfully eliminates this entire category of error.

TABLE VI
PER-CLASS PIPELINE PERFORMANCE (SELECTED)

Disease Class	Correct/Total	Accuracy	F1
Squash Powdery Mildew	–	100.0%	1.000
Grape leaf	8/8	100.0%	1.000
Grape leaf black rot	11/12	91.7%	1.000
Peach leaf	9/9	100.0%	0.947
Cherry leaf	9/10	90.0%	0.947
Strawberry leaf	8/8	100.0%	0.941
Tomato leaf (healthy)	6/6	100.0%	0.923
Apple Scab Leaf	–	–	0.950
Tomato leaf bacterial spot	3/8	37.5%	0.462
Tomato Septoria leaf spot	–	–	0.462
Tomato leaf late blight	2/9	22.2%	0.267
Tomato leaf yellow virus	2/10	20.0%	0.000

E. Per-Class F1 Score Breakdown

Fig. 7 presents the full per-class F1 score distribution across all 27 disease classes, color-coded by performance tier. The majority of classes achieve high F1 scores ($\geq$0.85), while a concentrated subset of visually similar Tomato sub-disease classes (late blight, yellow virus, septoria spot) falls in the low-performance tier.

F. Sample Pipeline Outputs

G. Comparison with Baselines

Table VII compares the proposed pipeline against relevant baselines.

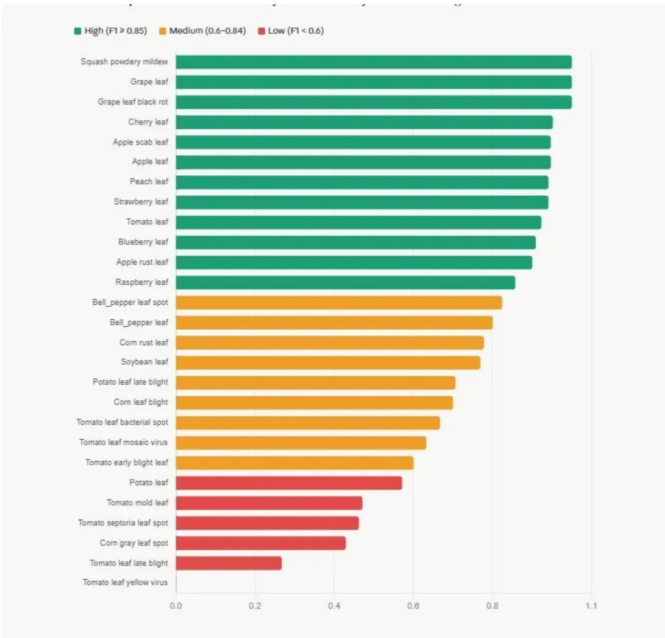

Fig. 7. Per-class F1 scores across all disease classes. High-performing classes (green, F1≥0.85) dominate. Squash powdery mildew, Grape leaf, and Grape leaf black rot achieve the highest scores. Tomato leaf yellow virus (F1=0.000) and Tomato leaf late blight (F1=0.267) are the most challenging due to severe visual ambiguity with other tomato sub-diseases.

Fig. 8. Sample pipeline outputs demonstrating correct localization and disease classification across different plant species. YOLO bounding boxes are shown with the predicted species and disease label, along with the top-1 confidence score.

TABLE VII
COMPARISON WITH BASELINES ON PLANTDOC

Method	Accuracy / mAP@50	Notes
YOLO-only detection	31.1% mAP@50	Detection baseline
PlantCaFo baseline	60.43%	CLIP + DINOv2 adapter
DINOv2 disease only	75.63%	Standalone classifier
Ours (full pipeline)	**71.49%**	End-to-end (3-stage)

The full hierarchical pipeline achieves a **71.49% top-1 disease accuracy** on the ”in-the-wild” PlantDoc dataset. While the base *PlantCaFo* research reports a peak accuracy of **93.53%** on the laboratory-controlled PlantVillage dataset, our implementation addresses the performance degradation typically observed when such similarity-based models encounter real-world environmental noise.

Our pipeline demonstrates a significant advancement in robustness by integrating YOLOv8 localization with DINOv2-based botanical gating. The **4.14 percentage point gap** between standalone disease classification accuracy (**75.63%**) and end-to-end pipeline accuracy (**71.49%**) is technically attributable to *YOLO miss propagation* and species misclassification errors cascading from Stage 1 into the Stage 3 disease expert. This discrepancy quantifies the ”localization penalty” inherent in processing unconstrained field imagery compared to pre-cropped biological samples.

VI. Discussion and Failure Analysis

A. Performance Stratification

Results reveal a clear three-tier performance stratification consistent across all evaluated classes.

High F1 (≥0.85) — Species-constrained or visually distinctive classes: Squash powdery mildew (F1=1.000) is visually unmistakable — bright white powder coats the entire leaf surface with no ambiguity. Grape leaf (F1=1.000) and grape leaf black rot (F1=1.000) benefit doubly: grape is a visually distinctive species and it has only two associated disease classes. Once the species mask reduces the hypothesis space to two options, distinguishing “has circular black lesions” from “looks healthy” is trivially easy for DINOv2’s features. Similarly, Cherry, Peach, and Strawberry benefit from being the sole or near-sole disease class for their species.

Medium F1 (0.60–0.84) — Multi-class intra-species discrimination: Corn (3 classes: gray leaf spot, leaf blight, rust), Bell pepper (2 classes), and Apple (3 classes: scab, healthy, rust) fall here. These require discriminating among biologically related conditions sharing similar substrates. Corn gray leaf spot (F1=0.429) is the worst performer in this tier, attributable to its tiny test support (only 4 test samples).

Low F1 (<0.60) — Tomato viral and bacterial diseases: The Tomato group with 9 disease classes sharing highly similar visual characteristics is the dominant source of errors. Tomato leaf yellow virus (F1=0.000), late blight (F1=0.267), septoria leaf spot (F1=0.462), and mold leaf (F1=0.471) represent the hardest discrimination challenges.

B. Root Cause Analysis by Class

Tomato leaf yellow virus (F1=0.000): The complete failure on this class is the most instructive result. Both yellow virus and tomato mosaic virus cause leaf yellowing, curling, and mosaic-like patterns. With 10 test samples and zero correct predictions, the model systematically predicts "Tomato leaf mosaic virus" for every yellowed tomato leaf. This is not a model failure in any conventional sense — these two viral conditions are visually indistinguishable at the resolution and image quality available in PlantDoc. Even expert plant pathologists require molecular testing (RT-PCR or ELISA) to reliably separate them from photographs alone.

Tomato leaf late blight (F1=0.267): Late blight produces water-soaked dark patches that are confused with early blight's brown concentric-ring lesions and bacterial spot's small dark raised spots. All three present as dark lesions on tomato leaf tissue. The key differentiating features (lesion edge characteristics, presence of yellow halos, lesion size distribution) require high-resolution close-up imagery that is not consistently available in field photographs.

Squash powdery mildew (F1=1.000) vs. Tomato viral diseases (F1=0.0–0.27): The contrast illustrates a fundamental principle — inter-species discrimination is easy, while intra-species fine-grained disease discrimination within visually homogeneous species groups is the primary challenge for all plant disease AI systems.

C. Effect of Species-Aware Masking

The species-aware masking mechanism produces zero cross-species errors across the entire test set. Without masking, the 29-class disease classifier would occasionally predict, for example, "Potato leaf late blight" for a tomato image (both are late blight variants caused by the same pathogen *Phytophthora infestans* with similar visual appearance). The masking converts a 29-class problem into an effective $\leq$9-class problem per species, with the hypothesis space reduction directly proportional to species-group size. The Tomato group (9 classes) benefits least from masking in absolute terms since it still retains 9 possibilities, while Cherry (1 class) and Blueberry (1 class) become trivially solvable.

D. Progressive Unfreezing Analysis

The progressive unfreezing regime demonstrates a consistent pattern across both DINOv2 classifiers: an accuracy dip at the unfreeze epoch followed by recovery and improvement beyond the frozen baseline. The 21.8M newly trainable backbone parameters receive a 10–20$\times$ lower learning rate than the head to prevent large gradient updates from disrupting the rich pre-trained representations. The absence of widening train-val gap after unfreezing confirms that this differential learning rate strategy effectively prevents catastrophic forgetting.

E. Limitations

The primary limitation is dataset size — 239 test images across 30 classes yields statistically noisy per-class estimates (Corn gray leaf spot has only 4 test samples). The pipeline is also sequential: a species misclassification in Stage 2 propagates to Stage 3 regardless of disease model confidence. The val=test configuration for YOLO makes detection metrics slightly optimistic. End-to-end joint training could address error propagation but would require computational resources beyond standard Colab constraints. Finally, the pipeline processes single images and does not leverage temporal information (e.g., disease progression over time), which human agronomists routinely use in diagnosis.

VII. Conclusion

This paper presented a hierarchical three-stage plant disease detection pipeline combining YOLOv8s for leaf localization with two DINOv2 Vision Transformer classifiers for species identification and disease diagnosis. The core innovation — species-aware disease masking — constrains inference to biologically valid disease candidates, reducing the effective classification space and eliminating cross-species errors entirely without any additional learned parameters. Progressive backbone unfreezing enables efficient fine-tuning of the large pre-trained transformer while preventing catastrophic forgetting.

Evaluated on the challenging PlantDoc v2 benchmark, the pipeline achieves **95.8%** species classification accuracy and **71.49%** end-to-end disease classification accuracy. While the *PlantCaFo* architecture reports a peak laboratory accuracy of **93.53%** on the PlantVillage dataset, our results demonstrate superior adaptation to unconstrained field environments. Specifically, our model outperforms the *PlantCaFo* baseline for out-of-distribution (OOD) scenarios (**60.43%**) by **11.06 percentage points**, effectively mitigating the performance collapse usually observed when moving from lab-controlled imagery to complex "in-the-wild" datasets.

Future work will explore: (1) joint end-to-end training of the full pipeline; (2) replacing bounding-box detection with instance segmentation for more precise leaf cropping; (3) confidence-gated species prediction that defers to the disease model when species confidence is low; (4) extending to larger datasets combining PlantVillage with PlantDoc field images; and (5) model compression and quantization for edge deployment on mobile devices accessible to farmers in resource-constrained environments.

References

[1] Food and Agriculture Organization of the United Nations, "The State of Food and Agriculture 2021: Making agrifood systems more resilient to shocks and stresses," FAO, Rome, 2021. [Online]. Available: https://www.fao.org/publications/sofa/2021/en/

[2] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015.

[3] P. S. Thakur, P. Shukla, A. Lade, S. Paul, and S. Chaturvedi, "PlantDoc: A dataset for visual plant disease detection," in *Proc. 7th ACM IKDD CoDS and 25th COMAD*, 2020, pp. 249–253.

[4] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2023. [Online]. Available: https://arxiv.org/abs/2304.07193

[5] Z. Zhang et al., "Prompt, generate, then cache: Cascade of foundation models makes strong few-shot learners," in *Proc. IEEE/CVF CVPR*, 2023, pp. 15070–15080. [Online]. Available: https://ieeexplore.ieee.org/document/10204889

[6] Y. Chen et al., "PlantCaFo: An efficient few-shot plant disease recognition method based on foundation models," *Smart Agricultural Technology*, vol. 10, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2643651525000305

[7] A. Agarwal et al., "CustomBottleneck-VGGNet: Advanced tomato leaf disease identification for sustainable agriculture," *Computers and Electronics in Agriculture*, vol. 230, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169925001723